

# Outsight Cloud Documentation

This document consolidates the **current state of Outsight Cloud (OC)**, including its architecture, technical documentation, core workflows, functionality overview, and the Airtable API surface used by the **Outsight Web Services (OWS)** project.

## 1. OC As-Is Presentation

What is Outsight Cloud (OC)?

**Outsight Cloud** is the web platform that lets customers and partners manage their **Outsight** LiDAR-related assets and workflows: organizations, sites, simulations, deployment forms, documentation, and file storage. It is implemented by the **OWS** repo (frontend app + API).

### As-Is Summary

| Aspect        | As-Is                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------|
| Product name  | Outsight Cloud                                                                                                 |
| Repo          | OWS (Outsight Web Services):<br>./app (frontend), ./api (backend)                                              |
| Auth          | Email + password; optional passwordless sign-in link; invitation-based signup; session in Redis or MemoryStore |
| Data store    | Airtable as primary database (single base app0FCTa3s476iDsI )                                                  |
| File storage  | S3-compatible (per region); simulations and site files (reference maps, deployment)                            |
| Deployment    | Docker Compose + Traefik; API on /api/ , app on / ; HTTPS + optional basic auth                                |
| Roles         | User, org-level permissions via roleAssignments ; Super Admin for admin/sessions/invites                       |
| Main entities | User → Organization (tree) → Site → Simulation; DeploymentForm; Documentations; Firmwares; Applications        |

### Key As-Is Characteristics

- **Single Airtable base** for all core data (users, organizations, sites, simulations, roles, docs, firmwares, applications, logs, deployment forms, lidar info).
- **Hierarchy**: Organizations can have parentOrganizationId (child orgs); Sites belong to an Organization; Simulations belong to a Site.
- **Permissions**: Stored in Airtable roleAssignments (userId, organizationId, permissions array, optional childorgs-scope); inheritance from parent org for child-org scope.
- **Hosting regions**: Sites/Organizations can have a bucketRegionId for S3; multiple regions supported via hostingRegions and S3 configurations.
- **Public simulations**: Some simulations are public and reachable without auth via /pubsimulation/simulations/:simulationId .

- **Documentation:** GitBook-backed; docs linked to organizations or “all users”; slug-based access under `/docs/:slug`.
  - **Staff notifications:** Key actions (auth, site/simulation creation, file ops) notify staff via Slack/Customer.io (`notifyStaff`).
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## 2. OC Technical Documentation

### Stack

- **Frontend:** Vue 3, Vite, TypeScript, Vue Router, Pinia, Tailwind.
- **Backend:** Node.js, Express, ES modules.
- **Session:** `express-session` with Redis store (production) or `MemoryStore` (dev).
- **Database:** `Airtable` (official `airtable` npm package).
- **Storage:** AWS S3-compatible (configurable endpoints per region).
- **Email:** Sendmail with Vue-based HTML templates (e.g. invitation, sendlink); from “Outsight Cloud [cloud@outsight.tech](mailto:cloud@outsight.tech)”.
- **Deployment:** Docker, Docker Compose, Traefik (HTTP/HTTPS, Let's Encrypt).

### Entry points

- **API:** `api/src/bin/www.js` → `api/src/app.js`; port `3001`; mounted at `/api/` by Traefik.
- **App:** SPA served on `/` (port `3000`); calls `/api/` for all data.

### Environment

- `PORT`, `NODE_ENV`, `BUILD_TAG`, `BUILD_TIMESTAMP`
- `REDIS_URL` (session store)
- `HOST` (used in emails and redirects)
- `CORS_WHITELIST` (optional)
- S3/config and Airtable API key are in code/env (see `api/src/database/airtable.js`, `api/src/services/s3/configurations.js`).

### Security

- Passwords: SHA-256 with fixed salt (`users.js`); validation (length, upper/lower/digits).
  - Sign-in link: time-based hash (12-hour window, hourly granularity).
  - Invitation link: time-based hash (30-day window, daily granularity).
  - Rate limiting on auth and sensitive endpoints (`limiter`).
  - Session cookie: `outsightsess`, `sameSite` strict, secure in production.
  - Admin routes protected by `userMustBeSuperAdmin`; organization/site/simulation by `userHasPermission` / `userHasRole` and resource ownership.
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## 3. Architecture Diagrams

### 3.1 High-level system architecture

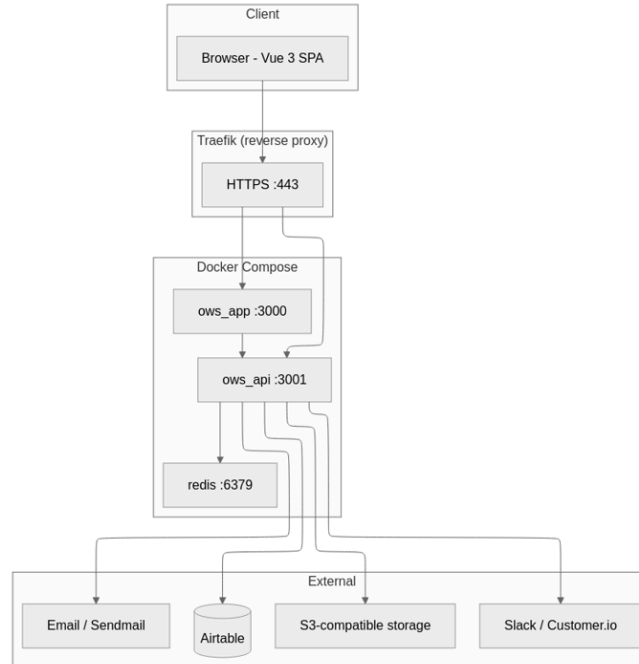
Code Mermaid for High-level system architecture

```
1 flowchart TB
2   subgraph Client
3     Browser[Browser - Vue 3 SPA]
4   end
5   subgraph Traefik["Traefik (reverse proxy)"]
6     HTTPS[HTTPS :443]
7   end
8   subgraph Docker["Docker Compose"]
9     OWS[ows_app :3000]
10    API[ows_api :3001]
11    Redis[redis :6379]
12  end
13  subgraph External
14    Airtable[Airtable]
15    S3[S3-compatible storage]
16    Mail[Email / Sendmail]
17    Notify[Slack / Customer.io]
18  end
19  Browser --> HTTPS
20  HTTPS --> OWS
21  HTTPS --> API
22  API --> Redis
23  API --> Airtable
```

```

24 API --> S3
25 API --> Mail
26 API --> Notify
27 OWS --> API
28

```



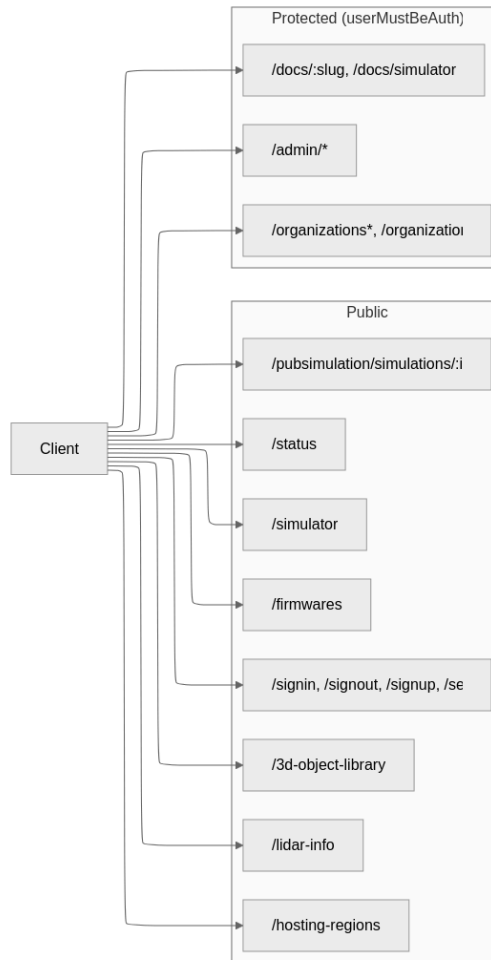
### 3.2 API route architecture

Code Mermaid for API route architecture

```

1 flowchart LR
2   subgraph Public
3     status["status[/status/]"]
4     auth["auth[/signin, /signout, /signup, /session, .../]"]
5     pubsim["pubsim[/pubsimulation/simulations/:id/]"]
6     lidar["lidar[/lidar-info/]"]
7     hosting["hosting[/hosting-regions/]"]
8     simulator["simulator[/simulator/]"]
9     firmwares["firmwares[/firmwares/]"]
10    object3d["object3d[/3d-object-library/]"]
11  end
12  subgraph Protected["Protected (userMustBeAuth)"]
13    admin["admin[/admin/*/]"]
14    orgs["orgs[/organizations*, /organizationbyslug/]"]
15    docs["docs[/docs/:slug, /docs/simulator/]"]
16  end
17  Client --> status
18  Client --> auth
19  Client --> pubsim
20  Client --> lidar
21  Client --> hosting
22  Client --> simulator
23  Client --> firmwares
24  Client --> object3d
25  Client --> admin
26  Client --> orgs
27  Client --> docs
28

```



### 3.3 Data model (Airtable + S3)

Code Mermaid for Data model

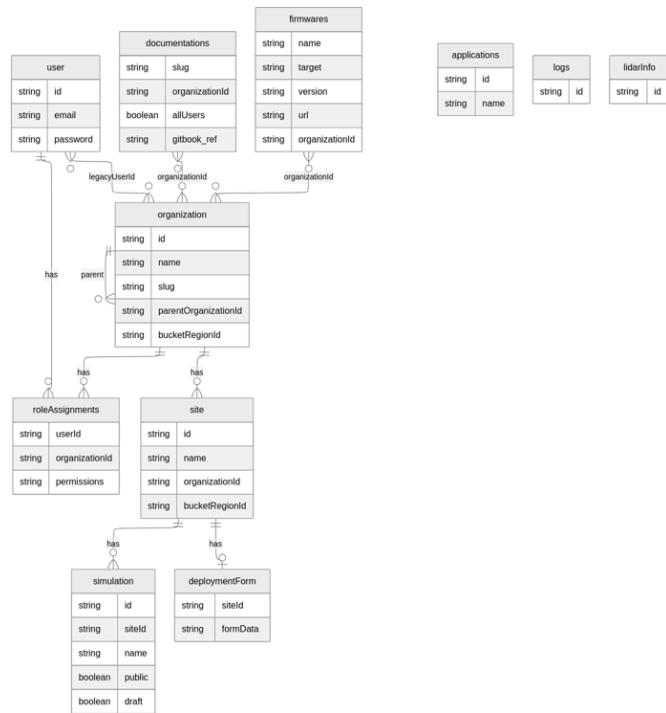
```

1 erDiagram
2   user ||--o{ roleAssignments : has
3   organization ||--o{ roleAssignments : has
4   organization ||--o{ organization : parent
5   organization ||--o{ site : has
6   site ||--o{ simulation : has
7   site ||--o{ deploymentForm : has
8   user }o--o{ organization : legacyUserId
9   documentations }o--o{ organization : organizationId
10  firmwares }o--o{ organization : organizationId
11  user {
12    string id
13    string email
14    string password
15  }
16  organization {
17    string id
18    string name
19    string slug
20    string parentOrganizationId
21    string bucketRegionId
22  }
23  site {
24    string id
25    string name
26    string organizationId
27    string bucketRegionId
28  }
29  simulation {
30    string id
31    string siteId
32    string name
33    boolean public
34    boolean draft
35  }
36  roleAssignments {
  
```

```

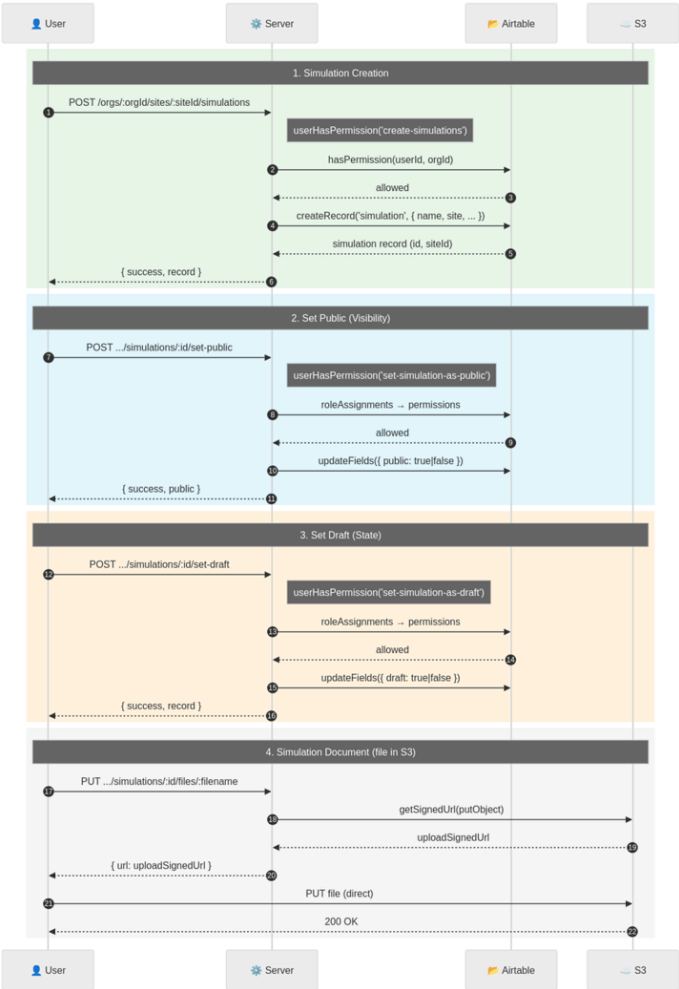
37     string userId
38     string organizationId
39     string permissions
40 }
41 deploymentForm {
42     string siteId
43     string formData
44 }
45 documentations {
46     string slug
47     string organizationId
48     boolean allUsers
49     string gitbook_ref
50 }
51 firmwares {
52     string name
53     string target
54     string version
55     string url
56     string organizationId
57 }
58 applications {
59     string id
60     string name
61 }
62 logs {
63     string id
64 }
65 lidarInfo {
66     string id
67 }
68 }

```

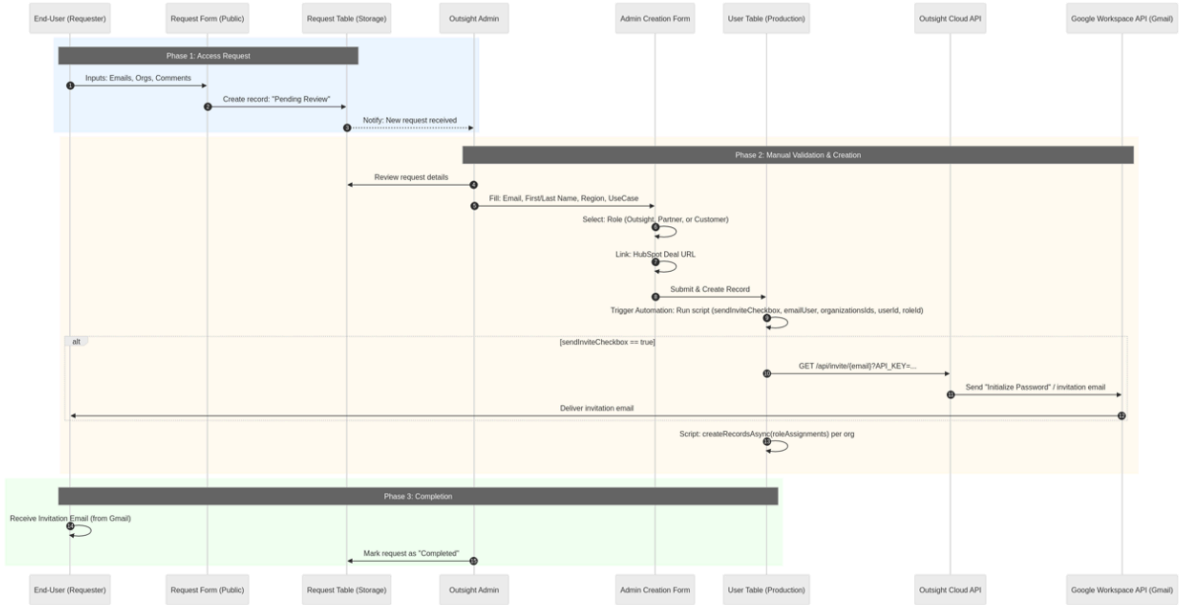


4. Flow Diagrams for Main Flows

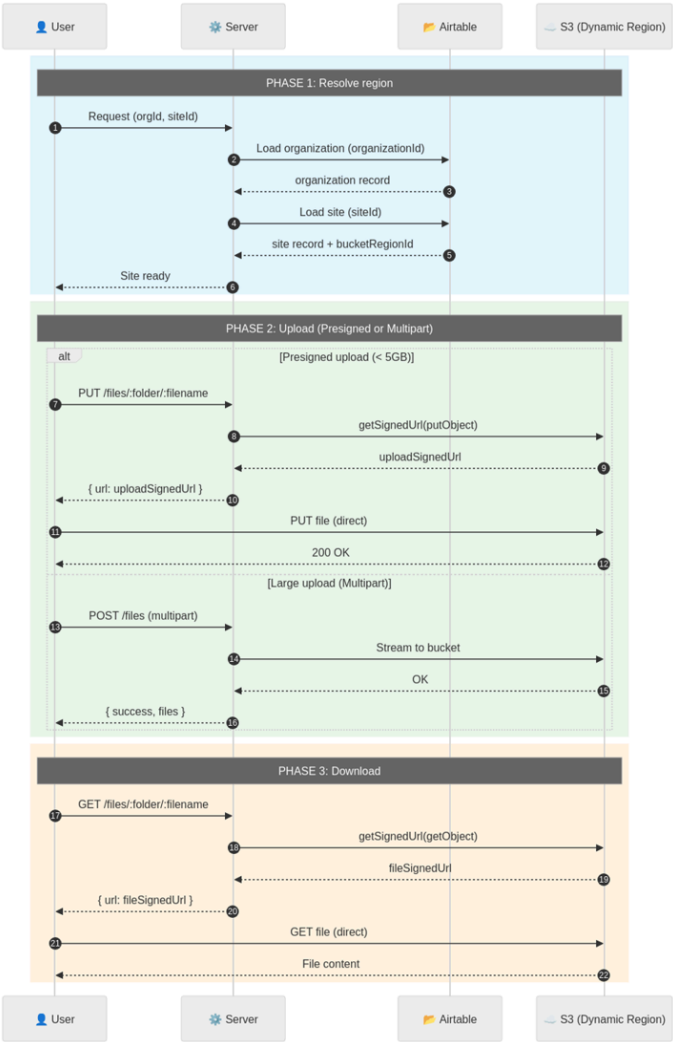
4.1 File upload and download



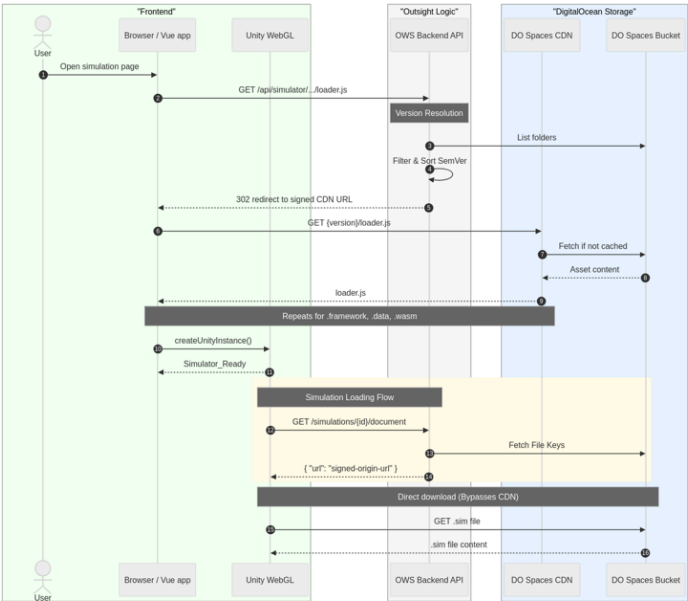
4.2 User registration



4.3 Simulation upload



4.4 Simulation Retrieval Flow





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## 5. Functionality List

### Authentication & account

- Sign-in with email + password.
- Sign-in with magic link (time-limited hash).
- Sign-out.
- Get current session ( GET /session ).
- Request account (form → staff notification; no auto-creation).
- Invitation link generation (superadmin or API key): GET /invitation-link/:email , GET /invite/:email (sends email).
- Signup with invitation hash + set password.
- Send “forgot password” link ( GET /sendlink/:email ).
- Set/change password (with current password or valid hash): POST /set-password .
- Superadmin: list all sessions ( GET /sessions ).

### Organizations

- List organizations for current user ( GET /organizations ).
- List all organizations flat (outsight-manager role) ( GET /organizations-flat ).
- Get organization by slug ( GET /organizationbyslug/:slug ).
- Get organization by id (via organizations?/:organizationId ).
- Get breadcrumb (org hierarchy).
- List child organizations ( GET /child-organizations ).
- Create child organization ( POST /organizations ) — note: missing imports in code for createOrganization and DateTime in organization/index.js .
- Create/delete top organization: not implemented (returns “Not implemented”).

### Sites

- List sites of an organization ( GET /organizations?/:organizationId/sites ).
- Get site ( GET /organizations?/:organizationId/sites?/:siteId ).
- Create site ( POST /organizations?/:organizationId/sites ).
- Update site (name, coordinates) ( POST /organizations?/:organizationId/sites?/:siteId ).
- Soft-delete site ( DELETE .../sites?/:siteId ).

### Simulations

- List simulations of a site ( GET .../sites?/:siteId/simulations ); draft visibility by permission.
- Get simulation ( GET .../simulations?/:simulationId ).
- Create simulation ( POST .../sites?/:siteId/simulations ).
- Update simulation (name, public) ( POST .../simulations?/:simulationId ).
- Set public ( POST .../set-public ).
- Set draft ( POST .../set-draft ).
- Mark document updated ( GET .../did-update-document ).
- Move simulation to another site ( POST .../move ).
- Duplicate simulation ( POST .../duplicate ).
- Soft-delete simulation ( DELETE .../simulations?/:simulationId ).
- Get simulation document URL ( GET .../document ).
- List/Get/Upload/Delete simulation files (presigned and multipart): GET /files , GET /files?/:filename , PUT /files?/:filename , DELETE /files?/:filename , deprecated PUT /save/:filename .

### Deployment form

- Get deployment form for site ( GET .../sites?/:siteId/deployment-form ).
- Save deployment form ( POST .../deployment-form ).

#### Site files (reference maps, deployment)

- List files by folder ( GET `.../sites?/:siteId/files?/:folder` ).
- Get download URL ( GET `.../files?/:folder/:filename` ).
- Get upload URL ( PUT `.../files?/:folder/:filename` ).
- Delete file ( DELETE `.../files?/:folder/:filename` ).
- Large file upload (multipart) ( POST `.../files?/:folder/:filename` ).

#### Applications (per site)

- Applications router mounted under site; see `routers/organization/site/applications.js` for list/download behavior.

#### Documentations

- Get doc by slug ( GET `/docs/:slug` ); returns name, description, and optional GitBook URL for “all users” docs.

#### Docs simulator

- Sub-routes under `/docs/simulator` for simulator-related docs.

#### Firmwares

- Public firmwares and organization-scoped firmwares via GET `/firmwares` (and org-scoped via organization route).

#### Hosting regions & 3D object library

- GET `/hosting-regions` for region list.
- 3D object library router for asset listing.

#### Public simulation (no auth)

- All GET (and allowed) routes under `/pubsimulation/simulations?/:simulationId/*` (document, files, etc.) for public simulations only.

#### Admin

- Export ( `/admin/export` ).
- Incomplete uploads ( `/admin/incomplete-uploads` ).
- Sessions ( `/admin/sessions` ).
- Identity check ( `/admin/is` ).

#### Status

- Health/status endpoint ( `/status` or similar).

#### Lidar info

- Public lidar info endpoint ( `/lidar-info` ).

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## 6. Airtable API Surface Overview

#### Base

- **Base ID:** `app0FCTa3s476iDsI`
- **API docs:** <https://airtable.com/app0FCTa3s476iDsI/api/docs>
- **Client:** `airtable` npm package; `base = at.base(owsBaseId)`.
- **Helpers:** `recordById(baseName, id)`, `createRecord(baseName, fields)`, `updateRecord(record, fields)`, `deleteRecord(baseName, id)` — note: `deleteRecord` has a bug: it uses literal `'baseName'` instead of the `baseName` parameter.

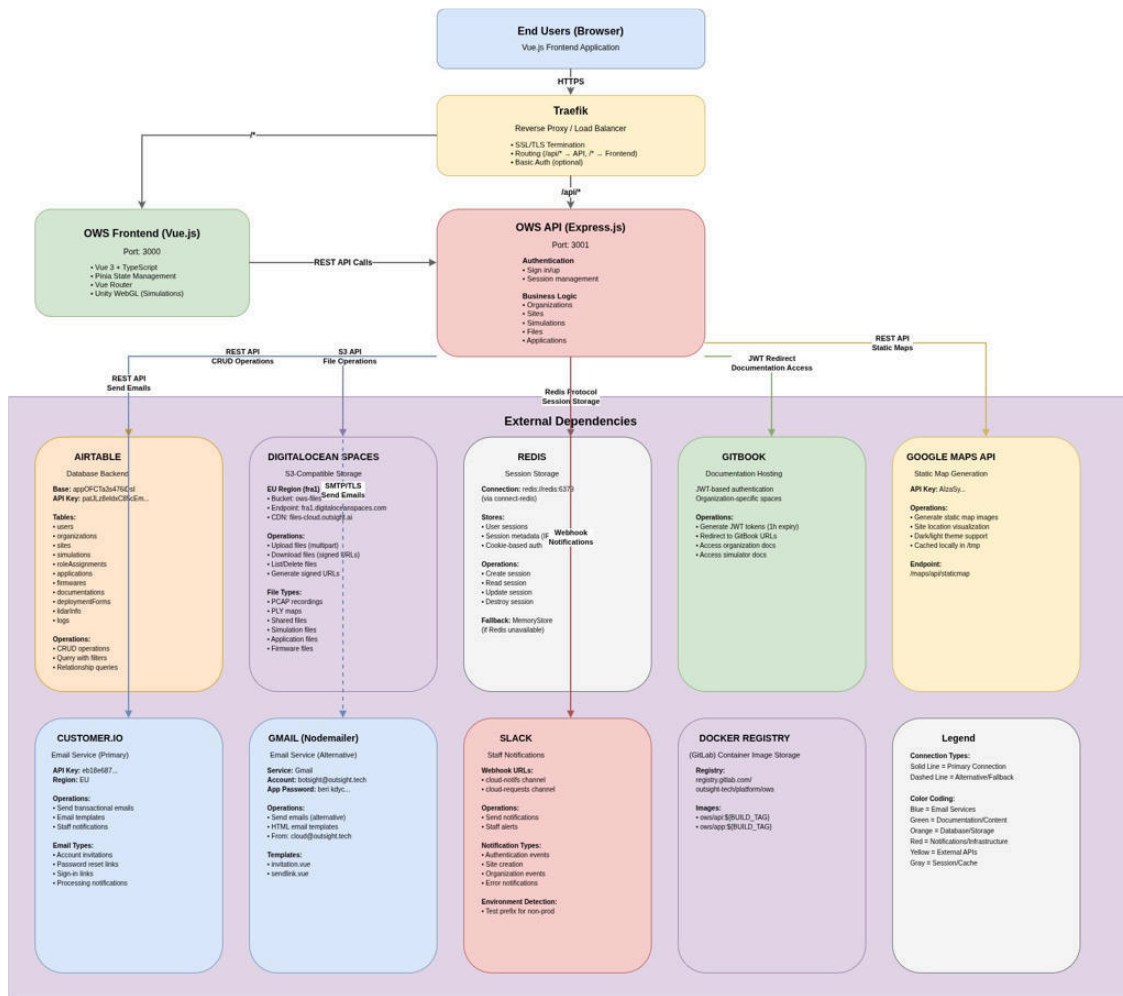
#### Tables and usage

| Table (base name) | Purpose | Read                    | Create | Update               | Delete | Main fields / notes |
|-------------------|---------|-------------------------|--------|----------------------|--------|---------------------|
| <b>user</b>       | Users   | by email (filter), view | —      | password, lastLogin, | —      | email, lc_email,    |

|                             |                         |                                                  |                        |                                                                                       |              |                                                                                                                                                                                                                     |
|-----------------------------|-------------------------|--------------------------------------------------|------------------------|---------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                         | api                                              |                        | lastIp                                                                                |              | password,<br>firstname,<br>lastname,<br>superadmin                                                                                                                                                                  |
| <b>organization</b>         | Organizations           | by id, by slug,<br>by userId,<br>child orgs, all | createOrgani<br>zation | —                                                                                     | deleteRecord | id, name,<br>slug,<br>cleanslug,<br>address, city,<br>zip, country,<br>parentOrgani<br>zationId,<br>userId,<br>bucketRegion<br>Id,<br>simulationEdi<br>torLidarsWhit<br>eList,<br>deletedAt                         |
| <b>site</b>                 | Sites                   | by id, by<br>organizationI<br>d                  | createSite             | updateFields<br>(name,<br>coordinates,<br>deletedAt)                                  | deleteRecord | id, name,<br>organizationI<br>d,<br>coordinates,<br>bucketRegion<br>Id,<br>firmwareUrls,<br>firmwareNam<br>es,<br>applications,<br>enableDeploy<br>mentForm,<br>deletedAt,<br>deletedBy,<br>createdAt,<br>createdBy |
| <b>simulation</b>           | Simulations             | by id, by<br>siteId                              | createSimulat<br>ion   | name, public,<br>draft, site,<br>documentUp<br>datedBy/At,<br>deletedAt,<br>deletedBy | deleteRecord | id, siteId,<br>name, public,<br>draft,<br>createdBy,<br>createdAt,<br>documentUp<br>datedBy,<br>documentUp<br>datedAt,<br>deletedAt,<br>deletedBy                                                                   |
| <b>roleAssignm<br/>ents</b> | User-org<br>permissions | by userId, by<br>(userId,<br>organizationI<br>d) | —                      | —                                                                                     | —            | userId,<br>organizationI<br>d,<br>permissions<br>(array),                                                                                                                                                           |

|                       |                          |                                        |                      |          |              |                                                                                         |
|-----------------------|--------------------------|----------------------------------------|----------------------|----------|--------------|-----------------------------------------------------------------------------------------|
|                       |                          |                                        |                      |          |              | childdorgs-<br>scope                                                                    |
| <b>logs</b>           | Logs                     | list (api view, 500)                   | createLog            | —        | —            | —                                                                                       |
| <b>lidarInfo</b>      | Lidar info (reference)   | list (api view)                        | —                    | —        | —            | —                                                                                       |
| <b>firmwares</b>      | Firmware versions        | by organizationId (SEARCH), public=1   | —                    | —        | —            | name, target, version, url, organizationId, public                                      |
| <b>documentations</b> | Doc entries for GitBook  | by slug, by organizationId, allUsers=1 | —                    | —        | —            | name, description, slug, organizationId, allUsers, gitbook_spaceUrl, gitbook_privateKey |
| <b>deploymentForm</b> | Deployment form per site | by siteId                              | createDeploymentForm | formData | deleteRecord | siteId, formData                                                                        |
| <b>applications</b>   | Applications catalog     | list, by id                            | —                    | —        | —            | —                                                                                       |

## 7. OC System Architecture Diagram



### Code drawio Architecture Diagram

```

1 <mxfile host="app.diagrams.net" modified="2024-01-
01T00:00:00.000Z" agent="5.0" version="21.0.0" etag="drawio-
diagram" type="device">
2 <diagram name="External Dependencies" id="external-
dependencies">
3 <mxGraphModel dx="2000" dy="1200" grid="1" gridSize="10"
guides="1" tooltips="1" connect="1" arrows="1" fold="1" page="1"
pageScale="1" pageWidth="2000" pageHeight="1600" math="0"
shadow="0">
4 <root>
5 <mxCell id="0" />
6 <mxCell id="1" parent="0" />
7
8 <!-- End Users -->
9 <mxCell id="users" value="<div style="text-align:
center; font-weight: bold; font-size: 16px; margin-bottom:
8px;">End Users (Browser)</div><div style="text-align: center; font-size: 12px;">Vue.js
Frontend Application</div>"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#dae8fc;strokeCo
lor=#6c8ebf;fontSize=12;spacingLeft=20;spacingRight=20;spacingTop=
15;spacingBottom=15;" vertex="1" parent="1">
10 <mxGeometry x="700" y="40" width="400" height="90"
as="geometry" />
11 </mxCell>
12
13 <!-- Traefik -->
14 <mxCell id="traefik" value="<div style="text-
align: center; font-weight: bold; font-size: 16px; margin-bottom:
10px;">Traefik</div><div style="text-align:
center; font-size: 12px; margin-bottom: 10px;">Reverse
Proxy / Load Balancer</div><div style="text-align:
left; font-size: 11px; padding-left: 10px;">• SSL/TLS
Termination<br>• Routing (/api/* -> API, /* ->
Frontend)<br>• Basic Auth (optional)</div>"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#fff2cc;strokeCo
lor=#d6b656;fontSize=12;spacingLeft=20;spacingRight=20;spacingTop=
15;spacingBottom=15;" vertex="1" parent="1">

```

```

15         <mxGeometry x="700" y="170" width="400" height="120"
as="geometry" />
16     </mxCell>
17
18     <!-- Frontend -->
19     <mxCell id="frontend" value="<div style="text-
align: center; font-weight: bold; font-size: 16px; margin-bottom:
10px;"&gt;OWS Frontend (Vue.js)&lt;/div&gt;&lt;div
style="text-align: center; font-size: 12px; margin-bottom:
10px;"&gt;&gt;Port: 3000&lt;/div&gt;&lt;div style="text-
align: left; font-size: 11px; padding-left: 10px;"&gt;&lt;• Vue 3
+ TypeScript&lt;br&gt;&lt;• Pinia State Management&lt;br&gt;&lt;• Vue
Router&lt;br&gt;&lt;• Unity WebGL (Simulations)&lt;/div&gt;&lt;div
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#d5e8d4;strokeCo
lor=#82b366;fontSize=12;spacingLeft=20;spacingRight=20;spacingTop=
15;spacingBottom=15;" vertex="1" parent="1">
20         <mxGeometry x="100" y="340" width="320" height="160"
as="geometry" />
21     </mxCell>
22
23     <!-- Backend API -->
24     <mxCell id="backend" value="<div style="text-
align: center; font-weight: bold; font-size: 16px; margin-bottom:
10px;"&gt;OWS API (Express.js)&lt;/div&gt;&lt;div
style="text-align: center; font-size: 12px; margin-bottom:
10px;"&gt;&gt;Port: 3001&lt;/div&gt;&lt;div style="text-
align: left; font-size: 11px; padding-left:
10px;"&gt;&lt;• &b&gt;Authentication&lt;/b&gt;&lt;br&gt;&lt;• Sign
in/up&lt;br&gt;&lt;• Session
management&lt;br&gt;&lt;• &b&gt;Business
Logic&lt;/b&gt;&lt;br&gt;&lt;• Organizations&lt;br&gt;&lt;•
Sites&lt;br&gt;&lt;• Simulations&lt;br&gt;&lt;• Files&lt;br&gt;&lt;•
Applications&lt;/div&gt;&lt;div
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#f8cecc;strokeCo
lor=#b85450;fontSize=12;spacingLeft=20;spacingRight=20;spacingTop=
15;spacingBottom=15;" vertex="1" parent="1">
25         <mxGeometry x="700" y="340" width="400" height="220"
as="geometry" />
26     </mxCell>
27
28     <!-- Connections: Users to Traefik -->
29     <mxCell id="edge1" value="HTTPS"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#666666;fontSize=12;
fontStyle=1;" edge="1" parent="1" source="users" target="traefik">
30         <mxGeometry relative="1" as="geometry" />
31     </mxCell>
32
33     <!-- Connections: Traefik to Frontend -->
34     <mxCell id="edge2" value="/*"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#666666;fontSize=12;
fontStyle=1;" edge="1" parent="1" source="traefik"
target="frontend">
35         <mxGeometry relative="1" as="geometry">
36             <Array as="points">
37                 <mxPoint x="600" y="230" />
38                 <mxPoint x="260" y="230" />
39             </Array>
40         </mxGeometry>
41     </mxCell>
42
43     <!-- Connections: Traefik to Backend -->
44     <mxCell id="edge3" value="/api/*"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#666666;fontSize=12;
fontStyle=1;" edge="1" parent="1" source="traefik"
target="backend">
45         <mxGeometry relative="1" as="geometry" />
46     </mxCell>
47
48     <!-- External Dependencies Group -->
49     <mxCell id="extGroup" value="<div style="text-
align: center; font-weight: bold; font-size: 18px; padding:
10px;"&gt;External Dependencies&lt;/div&gt;&lt;div
style="rounded=0;whiteSpace=wrap;html=1;fillColor=#e1d5e7;strokeCo
lor=#9673a6;fontStyle=1;fontSize=16;verticalAlign=top;align=center
;spacingLeft=10;spacingRight=10;spacingTop=10;spacingBottom=10;"
vertex="1" parent="1">
50         <mxGeometry x="60" y="620" width="1680" height="900"
as="geometry" />
51     </mxCell>
52
53     <!-- Airtable -->
54     <mxCell id="airtable" value="<div style="text-
align: center; font-weight: bold; font-size: 14px; margin-bottom:
10px;"&gt;AIRTABLE&lt;/div&gt;&lt;div style="text-align:
center; font-size: 11px; margin-bottom: 10px;"&gt;&lt;Database
Backend&lt;/div&gt;&lt;div style="text-align: left; font-
size: 10px; padding-left: 10px; line-height:
1.4;&quot;&gt;&lt;• &b&gt;Base&lt;/b&gt;&lt;br&gt;&lt;• API Key&lt;/b&gt;&lt;br&gt;&lt;• patJLz8e1dxC85cEm...&lt;br&gt;&lt;• &b&gt;Tables&lt;/b&gt;&lt;br&gt;&lt;• users&lt;br&gt;&lt;• organizations&lt;br&gt;&lt;•
sites&lt;br&gt;&lt;• simulations&lt;br&gt;&lt;• roleAssignments&lt;br&gt;&lt;•

```

```

applications&lt;br&gt;• firmwares&lt;br&gt;•
documentation&lt;br&gt;• deploymentForms&lt;br&gt;•
lidarInfo&lt;br&gt;•
logs&lt;br&gt;&lt;br&gt;&lt;b&gt;Operations:&lt;/b&gt;&lt;br&gt;•
CRUD operations&lt;br&gt;• Query with filters&lt;br&gt;•
Relationship queries&lt;br&gt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#ffe6cc;strokeCo
lor=#79b00;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
55 <mxGeometry x="100" y="680" width="280" height="380"
as="geometry" />
56 </mxCell>
57
58 <!-- DigitalOcean Spaces -->
59 <mxCell id="spaces" value="&lt;div style=&quot;text-align:
center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;DIGITALOCEAN SPACES&lt;/div&gt;&lt;div
style=&quot;text-align: center; font-size: 11px; margin-bottom:
10px;&quot;&gt;S3-Compatible Storage&lt;/div&gt;&lt;div
style=&quot;text-align: left; font-size: 10px; padding-left: 10px;
line-height: 1.4;&quot;&gt;&lt;b&gt;EU Region
(fra)&lt;/b&gt;&lt;br&gt;• Bucket: ows-files&lt;br&gt;• Endpoint:
fra.digitaloceanspaces.com&lt;br&gt;• CDN: files-
cloud.outsight.ai&lt;br&gt;&lt;br&gt;&lt;b&gt;Operations:&lt;/b&gt;
&lt;br&gt;• Upload files (multipart)&lt;br&gt;• Download files
(signed URLs)&lt;br&gt;• List/Delete files&lt;br&gt;• Generate
signed URLs&lt;br&gt;&lt;br&gt;&lt;b&gt;File
Types:&lt;/b&gt;&lt;br&gt;• PCAP recordings&lt;br&gt;• PLY
maps&lt;br&gt;• Shared files&lt;br&gt;• Simulation
files&lt;br&gt;• Application files&lt;br&gt;• Firmware
files&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#e1d5e7;strokeCo
lor=#9673a6;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
60 <mxGeometry x="420" y="680" width="280" height="380"
as="geometry" />
61 </mxCell>
62
63 <!-- Redis -->
64 <mxCell id="redis" value="&lt;div style=&quot;text-align:
center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;REDIS&lt;/div&gt;&lt;div style=&quot;text-align:
center; font-size: 11px; margin-bottom: 10px;&quot;&gt;Session
Storage&lt;/div&gt;&lt;div style=&quot;text-align: left; font-
size: 10px; padding-left: 10px; line-height:
1.4;&quot;&gt;&lt;b&gt;Connection:&lt;/b&gt;
redis://redis:6379&lt;br&gt;(via connect-
redis)&lt;br&gt;&lt;br&gt;&lt;b&gt;Stores:&lt;/b&gt;&lt;br&gt;•
User sessions&lt;br&gt;• Session metadata (IP,
timestamps)&lt;br&gt;• Cookie-based
auth&lt;br&gt;&lt;br&gt;&lt;b&gt;Operations:&lt;/b&gt;&lt;br&gt;•
Create session&lt;br&gt;• Read session&lt;br&gt;• Update
session&lt;br&gt;• Destroy
session&lt;br&gt;&lt;br&gt;&lt;b&gt;Fallback:&lt;/b&gt;
MemoryStore&lt;br&gt;(if Redis unavailable)&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#f5f5f5;strokeCo
lor=#666666;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
65 <mxGeometry x="740" y="680" width="280" height="380"
as="geometry" />
66 </mxCell>
67
68 <!-- GitBook -->
69 <mxCell id="gitbook" value="&lt;div style=&quot;text-
align: center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;GITBOOK&lt;/div&gt;&lt;div style=&quot;text-align:
center; font-size: 11px; margin-bottom:
10px;&quot;&gt;Documentation Hosting&lt;/div&gt;&lt;div
style=&quot;text-align: left; font-size: 10px; padding-left: 10px;
line-height: 1.4;&quot;&gt;&lt;b&gt;JWT-based
authentication&lt;br&gt;Organization-specific
spaces&lt;br&gt;&lt;br&gt;&lt;b&gt;Operations:&lt;/b&gt;
&lt;br&gt;• Generate JWT tokens (1h expiry)&lt;br&gt;• Redirect to GitBook
URLs&lt;br&gt;• Access organization docs&lt;br&gt;• Access
simulator docs&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#d5e8d4;strokeCo
lor=#82b366;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
70 <mxGeometry x="1060" y="680" width="280" height="380"
as="geometry" />
71 </mxCell>
72
73 <!-- Google Maps API -->
74 <mxCell id="googlemaps" value="&lt;div style=&quot;text-
align: center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;GOOGLE MAPS API&lt;/div&gt;&lt;div
style=&quot;text-align: center; font-size: 11px; margin-bottom:
10px;&quot;&gt;Static Map Generation&lt;/div&gt;&lt;div
style=&quot;text-align: left; font-size: 10px; padding-left: 10px;
line-height: 1.4;&quot;&gt;&lt;b&gt;API Key:&lt;/b&gt;
AIzaSy...&lt;br&gt;&lt;br&gt;&lt;b&gt;Operations:&lt;/b&gt;
&lt;br&gt;&lt;br&gt;&lt;b&gt;API Key:&lt;/b&gt;

```

```

gt;• Generate static map images&lt;br&gt;• Site location
visualization&lt;br&gt;• Dark/light theme support&lt;br&gt;•
Cached locally in
/tmp&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Endpoint:&lt;br&gt;&lt;br&gt;/map
s/api/staticmap&lt;br&gt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#fff2cc;strokeCo
lor=#d6b656;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
75 <mxGeometry x="1380" y="680" width="280" height="380"
as="geometry" />
76 </mxCell>
77
78 <!-- Customer.io -->
79 <mxCell id="customerio" value="&lt;div style=&quot;text-
align: center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;CUSTOMER.IO&lt;/div&gt;&lt;div style=&quot;text-
align: center; font-size: 11px; margin-bottom:
10px;&quot;&gt;&gt;Email Service (Primary)&lt;/div&gt;&lt;div
style=&quot;text-align: left; font-size: 10px; padding-left: 10px;
line-height: 1.4;&quot;&gt;&lt;br&gt;API Key:&lt;br&gt;
eb18e687...&lt;br&gt;&lt;br&gt;Region:&lt;br&gt;
EU&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Operations:&lt;br&gt;&lt;br&gt;•
Send transactional emails&lt;br&gt;• Email templates&lt;br&gt;•
Staff notifications&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Email
Types:&lt;br&gt;&lt;br&gt;• Account invitations&lt;br&gt;•
Password reset links&lt;br&gt;• Sign-in links&lt;br&gt;•
Processing notifications&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#dae8fc;strokeCo
lor=#6c8ebf;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
80 <mxGeometry x="100" y="1100" width="280" height="380"
as="geometry" />
81 </mxCell>
82
83 <!-- Gmail -->
84 <mxCell id="gmail" value="&lt;div style=&quot;text-align:
center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;&gt;GMAIL (Nodemailer)&lt;/div&gt;&lt;div
style=&quot;text-align: center; font-size: 11px; margin-bottom:
10px;&quot;&gt;&gt;Email Service (Alternative)&lt;/div&gt;&lt;div
style=&quot;text-align: left; font-size: 10px; padding-left: 10px;
line-height: 1.4;&quot;&gt;&lt;br&gt;Service:&lt;br&gt;
Gmail&lt;br&gt;&lt;br&gt;Account:&lt;br&gt;
botsight@outsight.tech&lt;br&gt;&lt;br&gt;App Password:&lt;br&gt;
beri
kduc...&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Operations:&lt;br&gt;&lt;br&gt;
• Send emails (alternative)&lt;br&gt;• HTML email
templates&lt;br&gt;• From:
cloud@outsight.tech&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Templates:&lt;br&gt;
t;&lt;br&gt;• invitation.vue&lt;br&gt;• sendlink.vue&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#dae8fc;strokeCo
lor=#6c8ebf;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
85 <mxGeometry x="420" y="1100" width="280" height="380"
as="geometry" />
86 </mxCell>
87
88 <!-- Slack -->
89 <mxCell id="slack" value="&lt;div style=&quot;text-align:
center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;&gt;SLACK&lt;/div&gt;&lt;div style=&quot;text-align:
center; font-size: 11px; margin-bottom: 10px;&quot;&gt;&gt;Staff
Notifications&lt;/div&gt;&lt;div style=&quot;text-align: left;
font-size: 10px; padding-left: 10px; line-height:
1.4;&quot;&gt;&lt;br&gt;&lt;br&gt;Webhook URLs:&lt;br&gt;&lt;br&gt;• cloud-
notifs channel&lt;br&gt;• cloud-requests
channel&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Operations:&lt;br&gt;&lt;br&gt;
• Send notifications&lt;br&gt;• Staff
alerts&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Notification
Types:&lt;br&gt;&lt;br&gt;• Authentication events&lt;br&gt;• Site
creation&lt;br&gt;• Organization events&lt;br&gt;• Error
notifications&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Environment
Detection:&lt;br&gt;&lt;br&gt;&lt;br&gt;• Test prefix for non-
prod&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#f8cecc;strokeCo
lor=#b85450;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
90 <mxGeometry x="740" y="1100" width="280" height="380"
as="geometry" />
91 </mxCell>
92
93 <!-- Docker Registry -->
94 <mxCell id="docker" value="&lt;div style=&quot;text-align:
center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;&gt;DOCKER REGISTRY&lt;/div&gt;&lt;div
style=&quot;text-align: center; font-size: 11px; margin-bottom:
10px;&quot;&gt;&gt;(GitLab) Container Image Storage&lt;/div&gt;&lt;div
style=&quot;text-align: left; font-size: 10px; padding-left: 10px;
line-height:
1.4;&quot;&gt;&lt;br&gt;&lt;br&gt;Registry:&lt;br&gt;&lt;br&gt;registry.gitlab
.com/&lt;br&gt;outsight-

```



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tech/platform/ows&lt;br&gt;&lt;br&gt;&lt;br&gt;&lt;br&gt;Images:&lt;/b&gt;&lt;br&gt;• ows/api:${BUILD_TAG}&lt;br&gt;•
ows/app:${BUILD_TAG}&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#e1d5e7;strokeCo
lor=#9673a6;fontSize=11;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
95   <mxGeometry x="1060" y="1100" width="280" height="380"
as="geometry" />
96   </mxCell>
97
98   <!-- Legend -->
99   <mxCell id="legend" value="&lt;div style=&quot;text-align:
center; font-weight: bold; font-size: 14px; margin-bottom:
10px;&quot;&gt;Legend&lt;/div&gt;&lt;div style=&quot;text-align:
left; font-size: 10px; padding-left: 10px; line-height:
1.6;&quot;&gt;&lt;br&gt;Connection Types:&lt;/b&gt;&lt;br&gt;Solid
Line = Primary Connection&lt;br&gt;Dashed Line =
Alternative/Fallback&lt;br&gt;&lt;br&gt;&lt;br&gt;Color
Coding:&lt;br&gt;&lt;br&gt;Blue = Email Services&lt;br&gt;Green =
Documentation/Content&lt;br&gt;Orange =
Database/Storage&lt;br&gt;Red =
Notifications/Infrastructure&lt;br&gt;Yellow = External
APIs&lt;br&gt;Gray = Session/Cache&lt;/div&gt;"
style="rounded=1;whiteSpace=wrap;html=1;fillColor=#f5f5f5;strokeCo
lor=#666666;fontSize=10;spacingLeft=15;spacingRight=15;spacingTop=
15;spacingBottom=15;align=left;verticalAlign=top;" vertex="1"
parent="1">
100  <mxGeometry x="1380" y="1100" width="280" height="380"
as="geometry" />
101  </mxCell>
102
103  <!-- Connections from Backend to External Services -
Better routing to avoid overlaps -->
104  <mxCell id="conn1" value="REST API&#xa;CRUD Operations"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#d79b00;endArrow=blo
ck;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="airtable">
105    <mxGeometry relative="1" as="geometry">
106      <Array as="points">
107        <mxPoint x="800" y="560" />
108        <mxPoint x="240" y="560" />
109      </Array>
110    </mxGeometry>
111  </mxCell>
112
113  <mxCell id="conn2" value="S3 API&#xa;File Operations"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#9673a6;endArrow=blo
ck;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="spaces">
114    <mxGeometry relative="1" as="geometry">
115      <Array as="points">
116        <mxPoint x="900" y="560" />
117        <mxPoint x="560" y="560" />
118      </Array>
119    </mxGeometry>
120  </mxCell>
121
122  <mxCell id="conn3" value="Redis Protocol&#xa;Session
Storage"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#666666;endArrow=blo
ck;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="redis">
123    <mxGeometry relative="1" as="geometry">
124      <Array as="points">
125        <mxPoint x="1000" y="560" />
126        <mxPoint x="880" y="560" />
127      </Array>
128    </mxGeometry>
129  </mxCell>
130
131  <mxCell id="conn4" value="JWT Redirect&#xa;Documentation
Access"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#82b366;endArrow=blo
ck;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="gitbook">
132    <mxGeometry relative="1" as="geometry">
133      <Array as="points">
134        <mxPoint x="1100" y="560" />
135        <mxPoint x="1200" y="560" />
136      </Array>
137    </mxGeometry>
138  </mxCell>
139
140  <mxCell id="conn5" value="REST API&#xa;Static Maps"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;je
ttySize=auto;html=1;strokeWidth=2;strokeColor=#d6b656;endArrow=blo
ck;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="googlemaps">
141    <mxGeometry relative="1" as="geometry">
142      <Array as="points">

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143         <mxPoint x="1100" y="540" />
144         <mxPoint x="1520" y="540" />
145     </Array>
146 </mxGeometry>
147 </mxCell>
148
149     <mxCell id="conn6" value="REST API&#xa;Send Emails"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;jettySize=auto;html=1;strokeWidth=2;strokeColor=#6c8ebf;endArrow=block;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="customerio">
150         <mxGeometry relative="1" as="geometry">
151             <Array as="points">
152                 <mxPoint x="800" y="560" />
153                 <mxPoint x="240" y="560" />
154                 <mxPoint x="240" y="1100" />
155             </Array>
156         </mxGeometry>
157     </mxCell>
158
159     <mxCell id="conn7" value="SMTP/TLS&#xa;Send Emails"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;jettySize=auto;html=1;strokeWidth=2;strokeColor=#6c8ebf;endArrow=block;endFill=1;dashed=1;fontSize=11;fontStyle=1;" edge="1"
parent="1" source="backend" target="gmail">
160         <mxGeometry relative="1" as="geometry">
161             <Array as="points">
162                 <mxPoint x="900" y="560" />
163                 <mxPoint x="560" y="560" />
164                 <mxPoint x="560" y="1100" />
165             </Array>
166         </mxGeometry>
167     </mxCell>
168
169     <mxCell id="conn8" value="Webhook&#xa;Notifications"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;jettySize=auto;html=1;strokeWidth=2;strokeColor=#b85450;endArrow=block;endFill=1;fontSize=11;fontStyle=1;" edge="1" parent="1"
source="backend" target="slack">
170         <mxGeometry relative="1" as="geometry">
171             <Array as="points">
172                 <mxPoint x="1000" y="560" />
173                 <mxPoint x="880" y="560" />
174                 <mxPoint x="880" y="1100" />
175             </Array>
176         </mxGeometry>
177     </mxCell>
178
179     <!-- Frontend to Backend connection -->
180     <mxCell id="conn9" value="REST API Calls"
style="edgeStyle=orthogonalEdgeStyle;rounded=0;orthogonalLoop=1;jettySize=auto;html=1;strokeWidth=2;strokeColor=#666666;endArrow=block;endFill=1;fontSize=12;fontStyle=1;" edge="1" parent="1"
source="frontend" target="backend">
181         <mxGeometry relative="1" as="geometry">
182             <Array as="points">
183                 <mxPoint x="420" y="420" />
184                 <mxPoint x="700" y="420" />
185             </Array>
186         </mxGeometry>
187     </mxCell>
188
189 </root>
190 </mxGraphModel>
191 </diagram>
192 </mxfile>

```